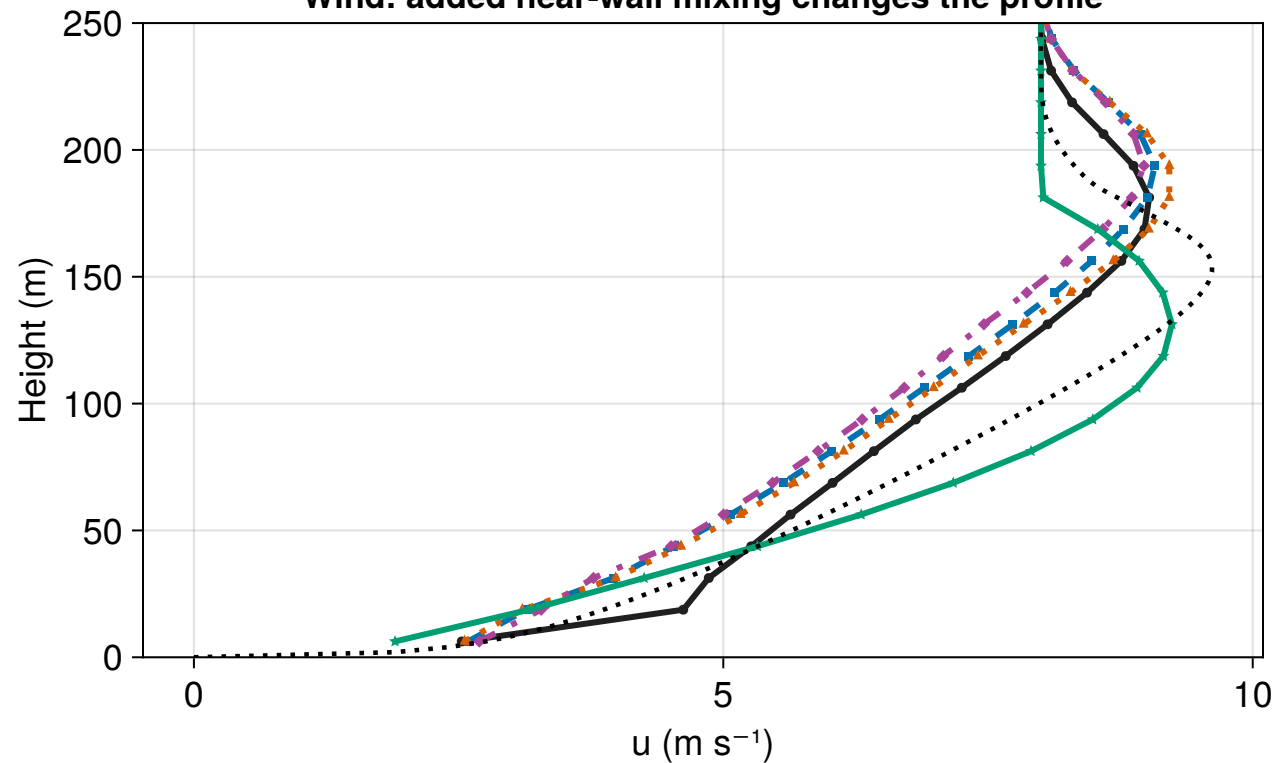


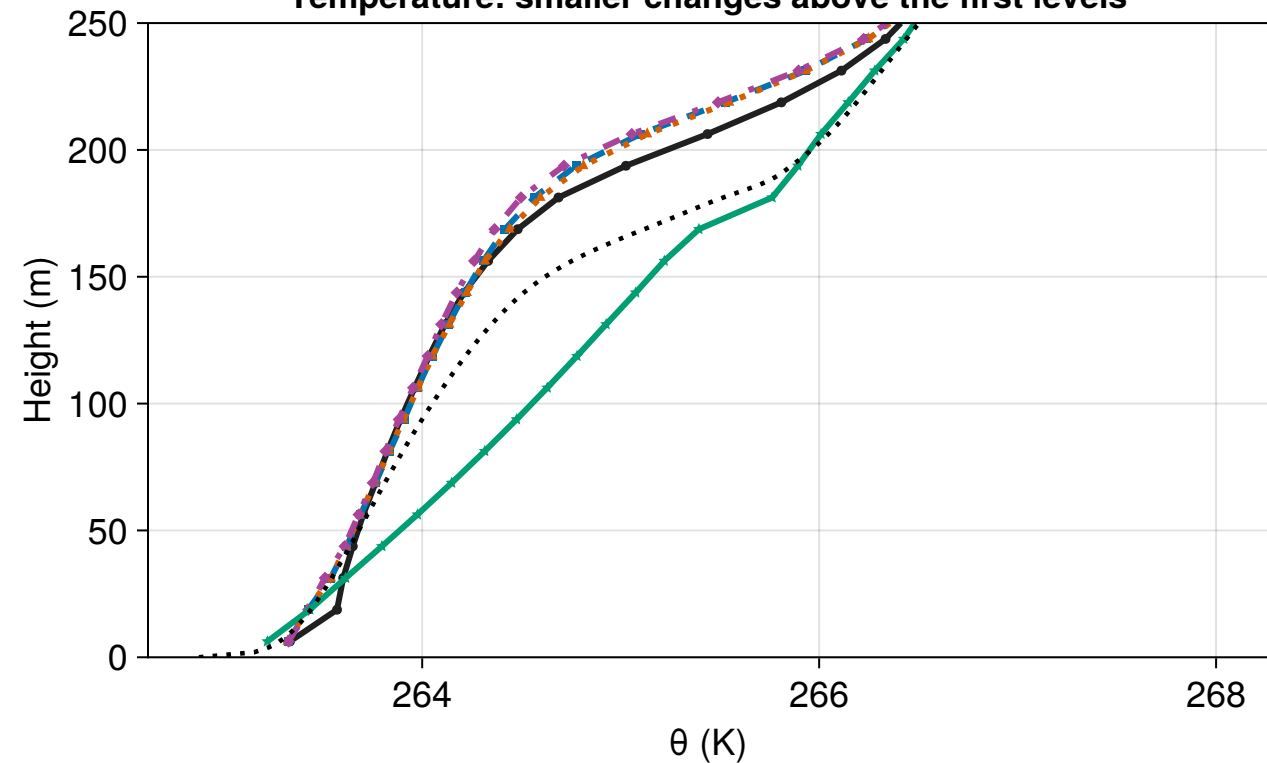
Surface-layer mixing versus coarse-grid Smagorinsky

GABLS1 · WENO9 · 12.5 m grid (32^3) · final hour, 8–9 h · four matched runs + earlier Smagorinsky

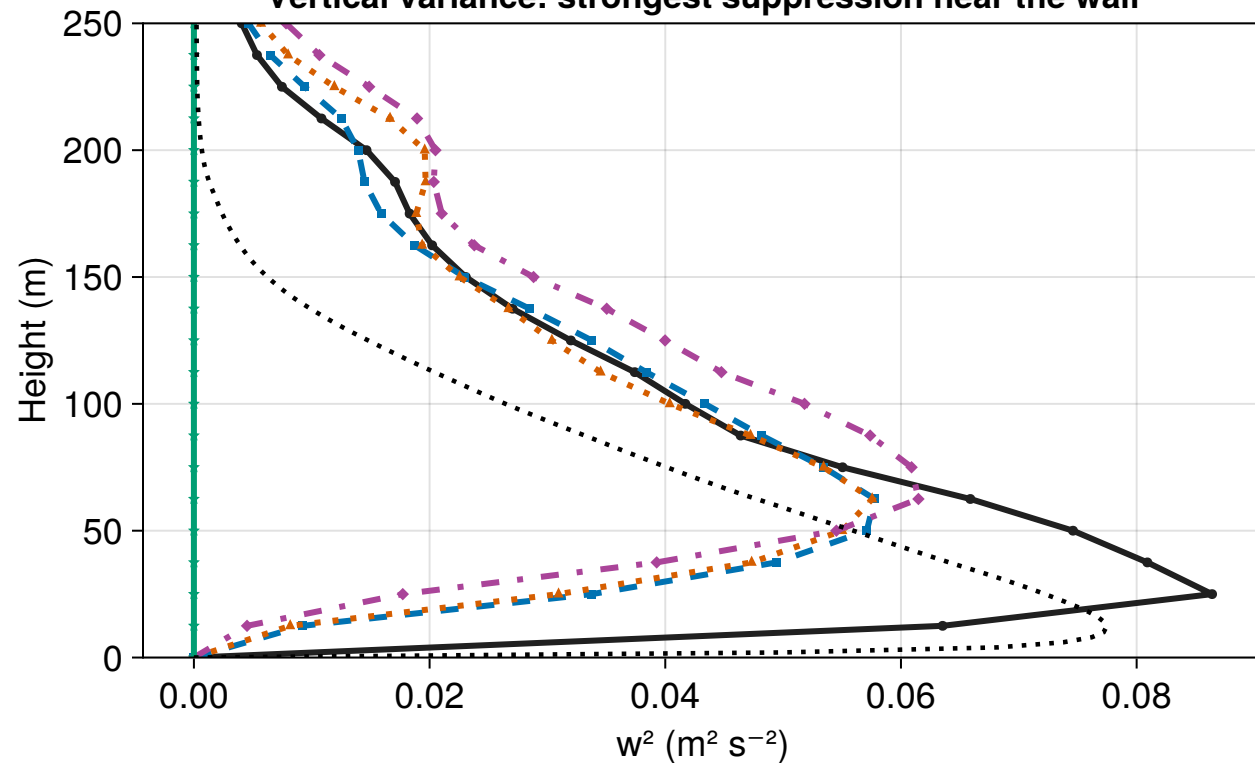
Wind: added near-wall mixing changes the profile



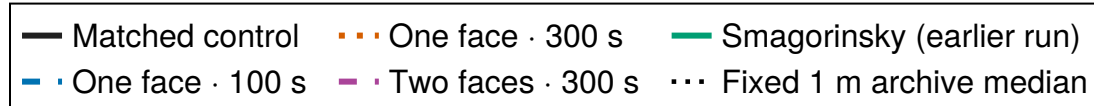
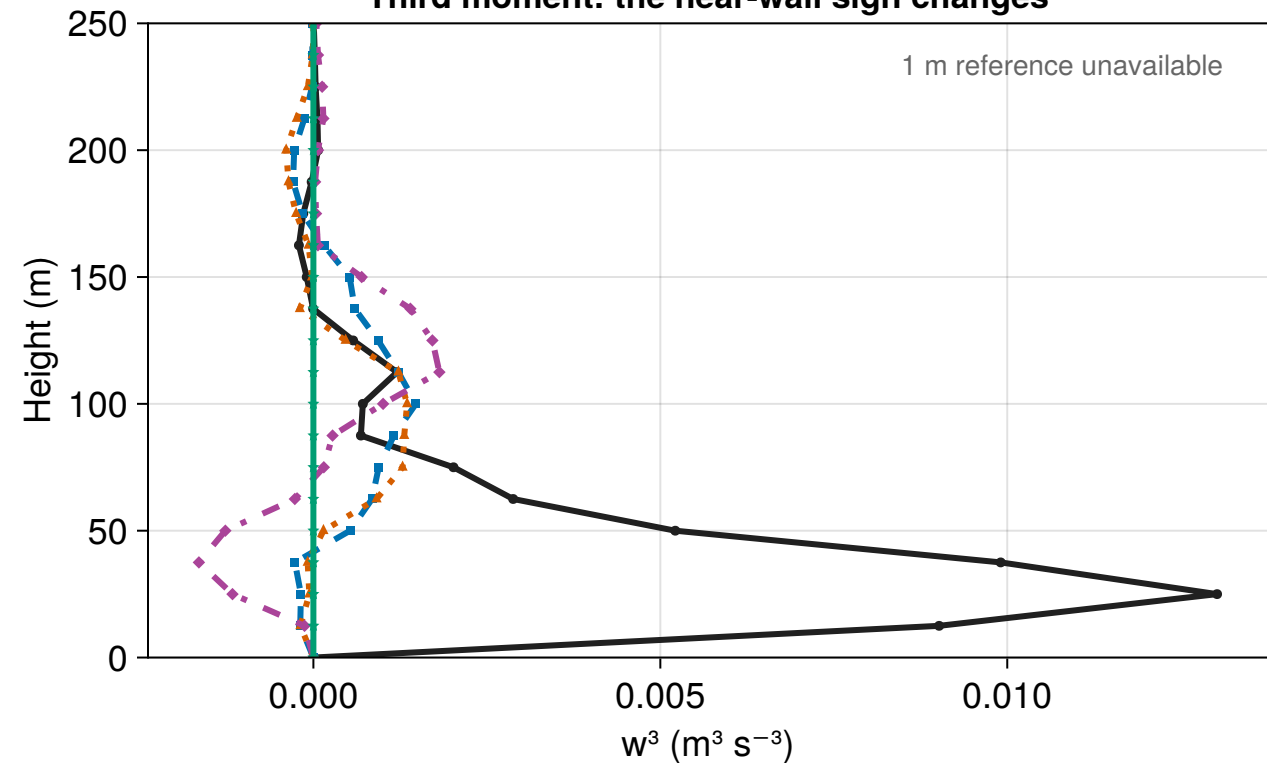
Temperature: smaller changes above the first levels



Vertical variance: strongest suppression near the wall



Third moment: the near-wall sign changes



Teal stars: earlier Smagorinsky source; its w^2 and w^3 nearly coincide with zero on these linear axes.
Black dotted: fixed 1 m LES median. Matched runs use the corrected, GPU-validated flux diagnostics.